Sharp Mixed-Entry Determinant Budgets for Uniform Transport under Weak MTW

Dr. Muralidhara Subbarao
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www.integralresearch.net
Stony Brook University
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Abstract

The assigned manuscript asserts uniform bi-Hölder optimal transport for the entire positive density-bounded class on a fixed compact weak-MTW manifold, including conjugate cut limits. We isolate its actual lower-jet determinant configuration. Retaining the center's isotropic slack and mixed block yields the least admissible transverse pole matrix, a sharp determinant minimum, and an exact compatibility condition when the scalar margin vanishes. A finite-error version allows controlled failure of annihilation and matrix inequalities, while explicitly requiring genuine lower-test remainders and physical Jacobian bounds. The source's already stronger quadratic endpoint jet gives a further conditional finite-time factor; it is not a new jet theorem or a uniform gain through the prefix-time limit. Exact examples exhibit the improvement and the necessity of mixed entries and continuum validation. Classical Schur and determinant tools are credited, and the geometric selection, conjugacy exclusion, uniform spectral comparison and global Hölder conclusion remain separate source inputs. Contribution assessment: C3/5, a sharp constrained determinant minimum and its explicit finite-error extension strengthen the source's local matrix interface without establishing a new global transport theorem.

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1 Problem, normalization and the complete source route

Let (M, g) be smooth, connected, compact and boundaryless, of dimension $n \geq 2$. Set $c(x, y) = d(x, y)^2/2$, and let $I(x)$ consist of velocities whose minimizing geodesic extends minimizing beyond duration one. Its closure G_x contains all minimizing velocities, including cut and conjugate ones. The weak-MTW condition of the source is

$$-\frac{3}{2}\partial_s^2\partial_t^2 c(\exp_x(t\xi), \exp_x(p + s\eta))\big|_{s=t=0} \geq 0, \qquad p \in I(x), \quad \xi \perp \eta.$$

No inequality at a nonsmooth cost pair is asserted by this formula. For fixed $0 < \lambda \leq \Lambda$, the source's density class consists of all measurable ρ_i with $\int_M \rho_i \, d\,\mathrm{vol} = 1$ and $\lambda \leq \rho_i \leq \Lambda$ almost everywhere. Its headline conclusion is one exponent and constant for every optimal map and inverse in that class on the fixed manifold [1]. No explicit exponent or uniformity over varying metrics is claimed there or here.

Use the negative-cost convention $u(x) = \sup_y\{-c(x, y) - v(y)\}$ and its dual relation for v. Then $u + c + v \geq 0$, with equality at contacts. Ordinary subgradients are identified using the Riemannian metric. Define

$$A_x(p) = D^2_{xx} c(x, \exp_x p), \qquad \sigma_x(p) = |\det(d\exp_x)_p|.$$

At a nonconjugate ordinary cut point, A_x means the specified smooth upper branch, not a Hessian of the nonsmooth minimum of all branches. All local matrices below are real symmetric forms in physical orthonormal tangent coordinates. An adjoint is therefore a transpose; nonorthogonal coordinate changes are explicitly accounted for.

1.1 Architecture and logical order

All nine assigned public text files were freshly read, including every mathematical module, the main file, bibliography and README. Their roles are as follows.

1. `geometry.tex` states the companion's density-free inputs: global ordinary-subgradient support, convex minimizing domains and gap sections, and intermediate $C^{1,1}$ regularization with Lipschitz poles. They are imported by this manuscript. Time-one injectivity is not an input for arbitrary potentials.

2. `spectral.tex` factors full Jacobi evolution through a fixed joining point. The broken-action Hessian satisfies $D(1 - \delta) = D(1) + \delta G_\delta$, with uniform positive bounds on G_δ. A bounded background in the scaled source Hessian is retained. Transverse MTW and a separating logarithmic scale then give an ordered singular-value comparison along extended affine segments.

3. `scales.tex` takes a section-oscillation supremum over the whole density class. Failure of every power bound gives logarithmic plateaus. The center template and cap are fixed before the outer curvature K, then scales are taken small. The parameter inequality holds for every positive active representation, not only a selected one.

4. `jets.tex` proves exact pole identities and endpoint lower jets, then excludes outer conjugacy using the original contact measure. The outer determinant bounds the center Hessian. Only afterward is the short-time minimizing map shown nonsingular, allowing the original almost-everywhere Jacobian equation at selected centers. Its final oblique Schur argument already retains all mixed entries.

5. `conclusion.tex` converts a common oscillation power bound into common map and inverse bounds. The reciprocal map comes from the reversed density pair, not from arbitrary-potential time-one injectivity.

Reading this route is not independent certification of every global input. The present proofs concern the precisely declared local configurations below.

2 The exact lower-jet interface already in the source

A lower Taylor jet (ζ, B) of f at zero means

$$f(h) \geq f(0) + \zeta \cdot h + \tfrac{1}{2} h^T B h + o(|h|^2).$$

Subtracting εI from B gives a genuine quadratic lower test on some sufficiently small neighborhood. A numerically recovered Hessian alone does not establish this inequality.

For $g = w_o^c$, $F = Q_t g$ and the fixed-pole coordinates $z(q) = \exp_x(tq)$, the source defines

$$L = D_q^2 \{c(x, z(q))/t - F(z(q))\}\big|_{q=p}.$$

If $p = \sum m_j p_j$ is an active barycenter and $E = \operatorname{span}\{p_j - p\}$, its exact first-minimum calculation gives $L \geq 0$, $LE = 0$ and, for $t < s < 1$,

$$H_{o,j} = m_j(L + LQ_sL), \qquad Q_s = \frac{A_x(tp)}{t} - \sum_j m_j \frac{A_x(sp_j)}{s}, \qquad Q_s|_{E^\perp} \geq \kappa_g(s-t)I. \tag{2.1}$$

This is already proved in source `jets.tex`167–301, not a new Ram result. The mixed split derivative is exactly $-I$ because the reference pole is fixed. The pullback jet exists without invertibility of $d\exp_x$; converting it to an endpoint test requires a nonconjugate branch. Jensen errors do not create an arbitrary linear error in this exact first-minimum jet.

The original contact relation obeys $\lambda \operatorname{vol}(A)/\Lambda \leq \operatorname{vol}(\mathcal{C}(A)) \leq \Lambda \operatorname{vol}(A)/\lambda$. Consequently a genuine lower test of an original potential at a before-cut $\log r$, with positive relative Hessian W, obeys

$$\sigma_y(r) \det W \leq \Lambda/\lambda. \tag{2.2}$$

This is not a density estimate for a modified envelope. At a selected good center, undoing the scalar modification and pulling back by the physical exponential differential B gives

$$V = lB^T\{\operatorname{Hess} v + A_y(r/l)\}B, \qquad \det V = l^n \sigma_x(p)^2 \frac{\text{original transport Jacobian}}{\sigma_y(r/l)}.$$

The squared Jacobian factor is essential. The source proves the needed good-center selection before this identity is used.

Its final matrix configuration, after all prescribed limits, has a unit vector e, $q = p \cdot e > 0$, and $u = p + de$ with $d > 0$, together with

$$L \geq 0, \quad Le = 0, \quad L \geq V + aI - \beta pp^T, \quad W \geq mL + \gamma uu^T, \qquad V > 0. \tag{2.3}$$

Here $a \asymp b/D > 0$, $\beta \asymp b/D^2$, $\gamma \asymp Kb/D^2$ and $m \geq m_* > 0$ in the source. The upper comparison $H \leq L$ has been combined with its center lower bound. We use W for the selected outer matrix to avoid conflicting indices. The source already proves $\det W \geq cK \det V$ by an oblique projection. Our increment determines the exact least L allowed by *all* of (2.3), rather than only its restriction to $p^\perp$.

3 A sharp mixed-entry lower envelope

The following statements are finite-dimensional theorems in the physical tangent space; they do not assert that every matrix example is realized by an optimal transport problem. Fix $|e| = 1$, $q = p \cdot e > 0$, and let Q have orthonormal columns spanning $p^\perp$. Put $J = [Q, e]$, an invertible but generally nonorthogonal matrix, and $A = V + aI > 0$. Write

$$J^T A J = \begin{pmatrix} A_0 & z \\ z^T & h \end{pmatrix}, \qquad \Delta = \beta q^2 - h. \tag{3.1}$$

Here a may be any real number for which $A > 0$; the source supplies $a > 0$, and a finite-error budget below may reduce it.

Theorem 3.1 (Least admissible pole matrix)**.** *Let $\beta > 0$. A symmetric $L \geq 0$ with $Le = 0$ and $L \geq A - \beta pp^T$ exists precisely in the following cases:*

(i) $\Delta > 0$, in which case every such matrix satisfies

$$L \geq L_* := J^{-T} \begin{pmatrix} B_* & 0 \\ 0 & 0 \end{pmatrix} J^{-1}, \qquad B_* = A_0 + \frac{zz^T}{\Delta}; \tag{3.2}$$

(ii) $\Delta = 0$ and $z = 0$, in which case the same statement holds with $B_* = A_0$.

In both cases L_* itself is admissible and has kernel exactly $\mathbb{R}e$. If $\Delta < 0$, or $\Delta = 0$ with $z \neq 0$, no such L exists.

Proof. The condition $Le = 0$ gives $J^T L J = \operatorname{diag}(B, 0)$. Since $Q^T p = 0$, the full center inequality is equivalent to

$$\begin{pmatrix} B - A_0 & -z \\ -z^T & \Delta \end{pmatrix} \geq 0. \tag{3.3}$$

Its last diagonal entry must be nonnegative. For $\Delta > 0$, complete the square on (v, t):

$$v^T(B - A_0)v - 2tz^T v + \Delta t^2 = v^T(B - A_0 - zz^T/\Delta)v + \Delta(t - z^T v/\Delta)^2.$$

Thus (3.3) is equivalent to $B \geq B_*$. If $\Delta = 0$, positivity for every real t forces $z = 0$, and the remaining condition is $B \geq A_0$. Since $A_0 > 0$, these conditions also guarantee $L \geq 0$. Congruence back to the physical frame proves both minimality and attainability, including the asserted kernel. □

Theorem 3.2 (Sharp constrained determinant minimum)**.** *Assume a feasible case of Theorem 3.1. Let $m, \gamma > 0$ and let u satisfy $w = u \cdot e \neq 0$. If $W \geq mL + \gamma uu^T$, then $W > 0$ and*

$$\frac{\det W}{\det V} \geq \mathcal{G} := \frac{m^{n-1}\gamma w^2 \det B_*}{\det(J^T V J)}. \tag{3.4}$$

For fixed $V, a, \beta, p, e, u, m, \gamma$, this lower bound is attained by $L = L_*$ and $W = mL_* + \gamma uu^T$.

Proof. In the J coordinates, write $J^T u = (v, w)$. The determinant of

$$\begin{pmatrix} mB_* & 0 \\ 0 & 0 \end{pmatrix} + \gamma \begin{pmatrix} v \\ w \end{pmatrix} \begin{pmatrix} v^T & w \end{pmatrix}$$

is $m^{n-1}\gamma w^2 \det B_*$: eliminate v by the triangular change $(x, t) \mapsto (x, t - v^T x/w)$ in its quadratic form. The same form is positive because its two nonnegative terms cannot vanish simultaneously on a nonzero vector. By Theorem 3.1, $W \geq mL_* + \gamma uu^T > 0$. Positive definite determinant is monotone in this order, as follows by conjugating with the smaller matrix's inverse square root. Dividing numerator and denominator by the same $\det(J)^2$ proves (3.4). Equality follows from the specified admissible pair. Sharpness is for these matrix premises, not for the global geometric theorem. □

3.1 Comparison with the source's retained Schur complement

Take $a = 0$ and put $S = h - z^T A_0^{-1} z > 0$. The source already obtains $m^{n-1}\gamma w^2/S$ before replacing S by an upper bound. For $\Delta > 0$, the determinant lemma instead gives the exact refinement

$$\mathcal{G} = \frac{m^{n-1}\gamma w^2}{S}\left(1 + \frac{z^T A_0^{-1} z}{\beta q^2 - h}\right) = \frac{m^{n-1}\gamma w^2}{S}\frac{\beta q^2 - S}{\beta q^2 - h}. \tag{3.5}$$

The additional factor is at least one and is strictly greater than one when $z \neq 0$. It comes from the mixed entries of the *center inequality*, not from restoring a Schur complement that the source forgot. Retaining $a > 0$ gives the stronger formula (3.4) with $A = V + aI$ and the unchanged denominator $\det(J^T V J)$. In the zero-margin feasible case there is no 0/0 formula: Theorems 3.1–3.2 apply directly.

The Gram matrix $G = J^T J$ satisfies $\det G = q^2/|p|^2$. Physical determinants are $\det(J^T W J)/\det G$; their ratio cancels this factor. In particular, the displayed oblique block B_* must not be treated as an orthonormal operator matrix when estimating its eigenvalues or numerical errors. The explicit cancellation, not an assumption that J is isometric, justifies (3.4).

4 A finite-error determinant contradiction

The next result weakens the exact annihilation and form inequalities at a finite stage. Its premises are certified inequalities between actual physical forms. They are not conclusions of fitting a finite number of function values.

Theorem 4.1 (Annihilation, center and outer error budget)**.** *Let $L \geq 0$, $V > 0$, $|e| = 1$, $p \cdot e > 0$, $u \cdot e \neq 0$ and $m, \gamma, \beta > 0$. Suppose symmetric H, W satisfy*

$$H \leq L + \eta I, \qquad H \geq V + aI - \beta pp^T - \delta I, \tag{4.1}$$

$$\|Le\| \leq r, \qquad W \geq mL + \gamma uu^T - \xi I, \tag{4.2}$$

where $\eta, \delta, r, \xi \geq 0$. Set

$$a_* = a - \eta - \delta - 2r, \qquad A = V + a_* I, \qquad \theta = \xi + 2mr.$$

Assume $A > 0$. Construct Δ, B_*, L_* using this A. An infeasible case in Theorem 3.1 contradicts the premises. In a feasible case put $W_* = mL_* + \gamma uu^T$, and suppose a physical lower bound $W_* \geq \omega I$ is certified with $\omega > \theta$. Then

$$W > 0, \qquad \frac{\det W}{\det V} \geq (1 - \theta/\omega)^n \frac{m^{n-1}\gamma (u \cdot e)^2 \det B_*}{\det(J^T V J)}. \tag{4.3}$$

Consequently an independent upper ratio $\det W/\det V \le R$ *is impossible if the right side is greater than* R.

Proof. Let $\Pi = I - ee^T$ and $L_0 = \Pi L \Pi$. It is positive semidefinite, $L_0 e = 0$, and

$$L - L_0 = ee^T L + \Pi L ee^T, \qquad \|L - L_0\| \le 2r.$$

The two center inequalities imply $L_0 \ge V + a_* I - \beta pp^T$. Theorem 3.1 applies to L_0, so $L_0 \ge L_*$ in a feasible case. The outer inequality gives $W \ge W_* - \theta I$. Since $W_* \ge \omega I$,

$$W \ge (1 - \theta/\omega) W_* > 0.$$

Determinant monotonicity and the exact determinant of W_* prove (4.3). A ratio upper bound must come from genuine geometric or analytic input, not from this matrix argument. □

Remark 4.2 (What the error terms mean). The source's exact pole matrix already has $Le = 0$, so its exact configuration uses $r = 0$. The theorem permits a separately validated approximate configuration, not a silent replacement of the source identity by noisy data. The terms η, δ include transformed Jensen and center-form errors; ξ is an outer lower-form defect. These must be physical operator bounds including mixed entries. The factor $(1 - \theta/\omega)^n$ is conservative. It preserves a finite strict contradiction without pretending that arbitrary fixed errors survive the source's vanishing-scale limits.

4.1 Matrix uncertainty and actual lower tests

If only an approximation $\widetilde{V}$ is available, with a certified $\left\|V - \widetilde{V}\right\| \le \varepsilon$ and $V_- := \widetilde{V} - \varepsilon I > 0$, replace V in the lower-envelope construction by V_-. The numerator bound remains valid, while

$$\det V \le \det V_+, \qquad V_+ = \widetilde{V} + \varepsilon I.$$

Thus replace the denominator in (4.3) by $\det(J^T V_+ J)$. This follows directly from Loewner order; it is preferable to dividing by an unverified computed determinant. For nonorthogonal coefficient data with Gram matrix G, the correct error norm is $\left\|G^{-1/2} E G^{-1/2}\right\|$ for a form error E.

A lower-test gate is different. Suppose a claimed quadratic expansion is actually justified by

$$f(h) \ge f(0) + \zeta \cdot h + \tfrac{1}{2} h^T B h - \tfrac{1}{2}\omega(|h|)|h|^2, \qquad \omega(r) \longrightarrow 0. \tag{4.4}$$

For each $\varepsilon > 0$, on a radius where $\omega \le \varepsilon$, the polynomial with Hessian $B - \varepsilon I$ is a lower test. If its physical relative Hessian stays positive and its gradient log remains before cut, the source's original contact-volume bound applies. Neither the modulus, the available radius, nor the before-cut condition follows from a computed Hessian. With merely a fixed defect $\omega \le \varepsilon_0$, the valid test loses $\varepsilon_0 I$; it cannot be restored by a limit.

Under a smooth endpoint chart $y = Y(h)$ and a function f, the full second-derivative formula is

$$D^2(f \circ Y) = DY^T \operatorname{Hess} f \, DY + \langle \nabla f, D^2 Y \rangle. \tag{4.5}$$

The gradient and second-chart term cannot be dropped. The source handles it through exact cost identities or controlled gradient errors. Likewise, a scalar modification $w = \varphi(v)$ contributes $\varphi' \operatorname{Hess} v + \varphi'' \nabla v \nabla v^T$. Its inverse chain rule is responsible for the rank-one term in the outer form. These are mathematical lower-test requirements, not extra matrix entries that can simply be estimated from a finite list of point samples.

5 Using, not rediscovering, the stronger source jet

Equation (2.1) is sometimes replaced by mL in the source's final argument. At a fixed pair of prefix and split times, its extra term can be retained quantitatively.

Proposition 5.1 (A conditional finite-time determinant factor)**.** *Assume the exact feasible configuration of Theorem 3.1, and suppose the actual outer lower form obeys*

$$W \geq m(L + LQL) + \gamma uu^T, \qquad Q|_{e^\perp} \geq \kappa I, \quad \kappa \geq 0.$$

Let F have orthonormal columns spanning $e^\perp$ and put $K_* = F^T L_* F > 0$. Then

$$\frac{\det W}{\det V} \geq \mathcal{G} \det(I + \kappa K_*). \tag{5.1}$$

This is sharp for the stated matrix inequalities when $L = L_*$, $Q = \kappa I$ and equality is taken in the outer bound.

Proof. As $Le = 0$, the image of L is in $e^\perp$ and $LQL \geq \kappa L^2$. Write $K = F^T LF \geq K_* > 0$. In the orthonormal basis $[F, e]$, the determinant of $m(L + \kappa L^2) + \gamma uu^T$ is

$$m^{n-1}\gamma(u \cdot e)^2 \det K \det(I + \kappa K).$$

Each ordered eigenvalue of K is at least its counterpart for K_*. The scalar function $x(1 + \kappa x)$ is increasing for $x > 0$; multiplication gives the claimed lower bound. This does not use the generally false operator-monotonicity of $X \mapsto X^2$. The relation $\det K_* = \det B_* / \det(J^T J)$ cancels the same coordinate volume factor as in Theorem 3.2. The equality choice proves sharpness in the matrix class. □

In the source, $\kappa = \kappa_g(s - t)$ is positive only at a fixed $t < s < 1$. It tends to zero if the allowed split interval shrinks as $t \uparrow 1$. No additional uniform-in-K global contradiction or better Hölder exponent is obtained merely by inserting (5.1). The result is useful for a finite-stage comparison whose time gap and lower tests are genuinely certified.

6 Worked configurations and negative controls

Example 6.1 (A strict mixed-entry gain). Take the physical Euclidean plane, $p = e = (0, 1)^T$, $Q = (1, 0)^T$, $u = 2e$, $m = 1/2$, $\gamma = 1$, $\beta = 2$, and

$$V = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \det V = 1.$$

For $a = 0$, $\Delta = 1$, $B_* = 3$ and $L_* = \operatorname{diag}(3, 0)$. The full center defect is

$$L_* - V + 2pp^T = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \geq 0.$$

The extremal outer matrix is $W_* = \operatorname{diag}(3/2, 4)$, and $\det W_* / \det V = 6$. The source's retained-Schur estimate gives 4, since $S = 1/2$; the final coarser use of $S \leq \beta q^2$ gives 1. Thus the new constrained minimum is not just that existing Schur estimate under a new name.

Retain $a = 1/4$. Now $\Delta = 3/4$ and $B_* = 9/4 + 4/3 = 43/12$. The sharp ratio becomes $43/6$. At $\kappa = 1/3$, Proposition 5.1 multiplies the original $a = 0$ ratio by $1 + \kappa B_* = 2$, giving 12. These are exact algebraic configurations, not claimed realized transport jets on a compact manifold.

Example 6.2 (An explicit finite-error margin). Use the same $V, p, e, u, m, \gamma, \beta$, but suppose the certified input bounds in Theorem 4.1 have

$$a = \tfrac{1}{2}, \quad \eta = \delta = r = \xi = \tfrac{1}{32}.$$

Then $a_* = 3/8$, $\theta = 1/16$, $\Delta = 5/8$ and $B_* = 19/8 + 8/5 = 159/40$. The physical matrix W_* is $\mathrm{diag}(159/80, 4)$, so the conservative choice $\omega = 1$ is valid. The certified ratio is at least

$$(15/16)^2 \frac{159}{20} = \frac{7155}{1024} > 6.$$

Thus an independent geometric upper ratio 6 would be contradicted at this finite error budget. This example is a budget calculation, not evidence that the source supplies these numerical constants. The premises without that upper ratio are consistent: $L = \mathrm{diag}(9/2, 0)$, $H = L$ and $W = mL + uu^T$ satisfy stronger center and outer inequalities with the declared error allowances.

Example 6.3 (Zero margin is a compatibility condition). With $p = e = (0, 1)^T$, take $A = I$ and $\beta = 1$. Then $\Delta = 0$, $z = 0$ and $L_* = \mathrm{diag}(1, 0)$ is feasible. If A instead has diagonal entries one and off-diagonal $t \neq 0$, $|t| < 1$, the same scalar margin is zero but no finite L is feasible. The determinant and positive diagonals of A do not repair the missing mixed compatibility. For $A_0 = 2, h = 1, z = 1$ and $\beta = 1 + \varepsilon$, the least transverse entry is $2 + 1/\varepsilon$, displaying the conditioning cost as the margin tends to zero.

Example 6.4 (Discarded mixed entries and a false lower test). If one deletes the off-diagonal entry in the center defect of Example 6.1, one might take $L = \mathrm{diag}(2, 0)$. But its actual defect is $\left(\begin{smallmatrix} 0 & -1 \\ -1 & 1 \end{smallmatrix}\right)$, whose determinant is -1, so the claimed center inequality fails. Similarly $\mathrm{diag}(1, \varepsilon)$ is positive but has no uniform positive lower bound as $\varepsilon \downarrow 0$; positive determinant alone cannot absorb a fixed outer error.

Even exact second-order differentiation must retain mixed terms. For $f(x, y) = \frac{1}{2}(x^2 + y^2) + 2xy$, both diagonal second derivatives are positive, whereas the full Hessian has eigenvalues $3, -1$. Finite values are weaker still. Given finitely many sampled points in a neighborhood of zero, choose a smooth negative bump supported in a small ball avoiding those points and zero. Adding it to a quadratic does not change any sampled value or any jet at zero but can invalidate a claimed lower support on the whole declared neighborhood. A validated remainder such as (4.4), not interpolation alone, is needed. This is not a counterexample to the source's genuine jets.

7 Original Ram proof interfaces and their limits

The original local-jet chapter [5] supplies a useful discipline for recovering differentiated quantities, not an independent transport regularity theorem. Its full Leibniz rows contain derivatives of variable coefficients as well as every retained derivative of the unknown field. The differentiated-row remainder includes discarded field derivatives and moving-boundary terms. In particular, it need not equal the derivative of the remainder of the undifferentiated row. This matters for (4.5): a chart varying with the endpoint is not a constant matrix, and its gradient-times-second-chart term belongs to the actual lower-jet calculation.

For a simple exact check, $(xf)' = f + xf'$; keeping only xf' loses f. Likewise, for $Y(h_1, h_2) = (h_1, h_2 + h_1^2)$ and $f(y) = y_2$, the endpoint Hessian is zero but $D^2(f \circ Y) = \mathrm{diag}(2, 0)$. It is entirely the second term in (4.5). For a nonlinear coefficient row, the same chapter differentiates the coefficient's dependence on every retained state derivative. Exact finite polynomial recovery is therefore useful only on its declared model class; off that class the inverse multiplies a separately controlled remainder.

No moment identity proves the lower-support condition (4.4) for an arbitrary nonsmooth transport potential.

The original calibration report [6] proves a related errors-in-design statement. If $A\vartheta = b$, the measured design and right side are $A + \Delta A, b + e$, and $\sigma_{\min}(A + \Delta A) \geq s > 0$, then the least-squares minimizer satisfies

$$\left\| \widehat{\vartheta} - \vartheta \right\| \leq \frac{2}{s}(\|e\| + \|\Delta A\| \, \|\vartheta\|).$$

Indeed its measured residual is no larger than the residual at ϑ, and the triangle inequality bounds $(A + \Delta A)(\widehat{\vartheta} - \vartheta)$ by twice that latter residual. For an exactly solved square system the factor is one. The compactly supported weak atoms there require the full formal-adjoint product derivatives. These facts justify separating matrix uncertainty, continuum remainder and singular-value conditioning; they do not supply a missing lower-test modulus or a before-cut margin.

The physical-metric chapter [7] fixes the interpretation of nonorthogonal coefficients. A form E in a frame with Gram matrix G has physical operator norm $\left\| G^{-1/2} E G^{-1/2} \right\|$. This is exactly why the oblique J in Section 3 is used by congruence, with its determinant cancelled, rather than declared orthogonal. The same chapter separates finite compression from the omitted continuum residual. The present contribution is the application to the source's rank-one-defective center inequality, including its least feasible matrix and quantitative error losses. The classical Schur, Gram and determinant identities themselves are not claimed as new.

8 Advantages, costs and conditional extensions

The sharp envelope answers a concrete question not answered by the source's final coarse cK estimate: how much transverse pole curvature is forced by a given center form, including its mixed entries? It provides an attained minimum, an exact zero-margin compatibility gate, and a finite strict-contradiction budget. It does not require discarding the positive isotropic slack. At a fixed split time, the source's stronger quadratic jet gives the additional factor in Proposition 5.1 without a false operator-monotonicity argument.

These benefits have explicit costs. The small denominator Δ can amplify uncertain mixed entries. The factor $\omega - \theta$ must remain positive in the physical norm. A verified matrix approximation does not certify the radius of a lower support, branch selection, or the two original density laws. A coordinate Jacobian cannot be inverted at conjugacy merely because a finite Hessian is nonsingular. The robust theorem assumes all these missing ingredients separately. It is also conservative in replacing annihilation error by $2r$ and outer error by an isotropic loss; it makes no measured computational speed claim.

8.1 What remains global

The report does not replace the following dependencies of the assigned manuscript: companion global ordinary-subgradient support and convex injectivity domains; uniform Jacobi bounds through the correct joining factorization; logarithmic spectral ordering; plateau selection over the entire measurable density class; genuine Jensen contacts; removal of conjugate outer contacts by original contact volume; and nonsingular short-time pullback to original good centers. The source's ordered choice of constants remains necessary: geometric constants and the center template are fixed before K, with the small-scale and prefix-time choices afterward. A gain depending badly on K, a vanishing Δ, or a shrinking time gap cannot simply be inserted as a uniform improvement of its final Hölder modulus.

The quadratic refinement is therefore a conditional finite-time extension, not a new global endpoint jet. The exact sharp minimum and error theorem are locally proved here. The headline uniform bi-Hölder conclusion remains attributed to the source and is not certified by this report or its fixtures. M690 concerned a distinct singular Jacobi endpoint problem; no previous report's proofs or test counts are used as fresh evidence for M691.

Future work could combine certified lower-jet remainders with separate interval bounds on the true exponential differentials, then evaluate the full physical matrices of a concrete transport example. An anisotropic outer-error form could replace the scalar θI: one would retain $\det(W_* - E)$ after separately proving $W_* - E > 0$. To obtain a new global exponent instead would require uniform control of all relevant margins along the source's full scale selection. None of those quantitative geometric inputs is asserted here.

9 Current primary context and evidence

Figalli–Kim–McCann [2] establish Hölder continuity and injectivity under their weak-MTW, strong cost-convexity and density assumptions. The freshly inspected arXiv abstract and version record are v1, July 6, 2011; their proof was not independently audited here. Warren's quantitative stability result [3] has arXiv v1 of July 11, 2025 and a June 23, 2026 journal publication. The inspected publisher introduction and Theorem 1 use bounded C^2 log-densities and small transport cost to control the map derivative. This is a different regime from the assigned source's full measurable density-bounded class. No new proof of either result is claimed.

On October 9, 2026 the public path-history endpoint reported the source's latest path commit as the initial commit of October 6, `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. This is live metadata, not independent validation of remote bytes or mathematical correctness. The full family-360 repository declaration [4] was freshly read: it declares the uniform density-class result, while explicitly limiting its arbitrary-potential local injectivity declaration to finite families. This declaration is neither a Lean build nor a proof or axiom audit performed here.

9.1 Literal scope and reproducible checks

The assigned public scope is nine text files, 3443 lines, all displayed in bounded untruncated intervals. The local evidence records hashes persisted before the first scientific display and matching endpoint hashes afterward. Fresh private proof reading comprises 803 lines in three originals: the local-jet chapter, lines 513–615, 747–856, 897–937 and 1211–1308; the calibration report, lines 167–270 and 367–503; and all 210 lines of the physical-metric chapter. The complementary 2483 private lines were not read. A partial statement at the end of a displayed interval is not claimed as a complete proof; external references and included files not separately displayed remain uninspected. There is no whole-corpus audit.

The owned executable checks use rational matrices and symbolic derivatives, including oblique frames, attained determinant minima, zero-margin infeasibility, physical coordinate cancellation, the finite error example, mixed chain rules and negative controls. They test finite identities and inequalities only. They do not certify that a manufactured configuration is realized by a transport map or establish any global regularity claim. Parent checks are separate and are not included in the owned counts. The final owned script passed 100 exact cases and 1251 counted assertions, with zero floating diagnostics. Its result receipt records the script SHA-256 and execution time; the paper's examples use the same rational data.

10 Conclusion

Keeping the full center block produces an exact least feasible pole matrix rather than only a transverse restriction. Its attained determinant minimum strictly improves the source's retained-Schur estimate when the mixed entries are nonzero, and the finite-error extension exposes the necessary positivity and remainder margins. These constitute a notable local, source-specific contribution using classical algebra. They neither replace the geometric lower-test construction nor resolve the global uniformity and conjugacy steps.

References

[1] OpenAI mathematics source repository. *Uniform Bi-Hölder Transport from Weak MTW*, September 25, 2026. Pinned local manuscript, sections `geometry`, `spectral`, `scales`, `jets` and `conclusion`. Public source directory. Full assigned text read; global claims attributed.

[2] A. Figalli, Y.-H. Kim and R. J. McCann. *Hölder continuity and injectivity of optimal maps*. arXiv:1107.1014, v1 (2011). Abstract and version record inspected; proof not audited here.

[3] M. Warren. *A quantitative stability result for regularity of optimal transport on compact manifolds*. arXiv:2507.09035, v1 (2025); *Calc. Var. PDE* 65, article 214 (2026), journal version. Abstract, metadata, publisher introduction and Theorem 1 inspected.

[4] OpenAI mathematics repository. Family 360 scope declaration. All 20 displayed lines read October 9, 2026. Repository declaration only; no formal checker execution or proof audit.

[5] M. Subbarao. Original local-jet proofs, *RamNonlinearIE-PDE*, TR1-lin-nonlin-rev2, `sec-03-jets.tex`. Confidential local original, the intervals specified in Section 9.1; other portions unread.

[6] M. Subbarao. Original robust calibration and inversion report, TR207, `TR207-rmt-robust-calibration-inversion.tex`. Confidential local original, lines 167–270 and 367–503.

[7] M. Subbarao. *RamCalBooks*, volume 21, `ch01d-physical-metrics.tex`. Confidential local original, all 210 lines freshly read; external dependencies not thereby audited.